BHARATI MUDIGOUDRA AI Engineer

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SUMMARY

AI Engineer with hands-on experience building applications powered by LLMs, RAG pipelines, agentic workflows, and computer vision systems through internships and independent projects. Skilled in Python, FastAPI, LangChain, LangGraph, and vector databases; lead project ZeroQueue is containerized with Docker and runs pytest on every push via GitHub Actions. Also brings professional experience supporting US insurance clients - a strong fit for AI roles in insurance technology.

SKILLS

GenAI & LLM Engineering - LLMs, RAG, Prompt Engineering, AI Agents, Agentic Workflows, Embeddings, Vector Search, Reranking (CrossEncoder), NL2SQL, ReAct, MCP, LoRA and VLMs (basic knowledge)
AI Frameworks & Backend - Python, FastAPI, Flask, Django, LangChain, LangGraph, LlamaIndex, PyTorch, Scikit-learn
Machine Learning & Data - Feature Engineering, Model Selection, Classification, Regression, Clustering, Ensemble Learning, Model Evaluation, Pandas, NumPy
Computer Vision & NLP - Object Detection, Image Classification, Segmentation, YOLO, Transformers, LSTM, RNN
Data & Vector Infrastructure - SQL, MySQL, PostgreSQL, MongoDB, Pinecone, Weaviate, ChromaDB, FAISS
Cloud & MLOps - AWS, Docker, Git, GitHub Actions, CI/CD, pytest
Evaluation & Monitoring - LangSmith, Ragas, DeepEval

PROJECTS

Ask OLake - Documentation Q&A Assistant
- Built a Python and FastAPI assistant over the OLake data-ingestion docs at the Hacktoberfest Hack Day Bengaluru x OLake hackathon, answering setup, sync and Iceberg table-maintenance questions with numbered steps and cited sources.
- Implemented TF-IDF passage retrieval over 121 pages (698 passages), with relevance and evidence checks to reduce unsupported answers and exact quote checking for optional LLM answers.
- Shipped a public POST /ask API and an embeddable compact chat widget with source links that jump to the doc section when a matching section is identified; deployed on Vercel and tested with 188 pytest checks.

Tech Stack: Python, FastAPI, scikit-learn TF-IDF, JavaScript, Vercel, pytest

ZeroQueue - AI Gift-Card Support Copilot
- Co-built an AI gift-card support copilot with a teammate during the HiDevs YC Fall 2026 "Moss Zero-Latency Builder Sprint" hackathon.
- Contributed the Python and FastAPI backend with a RAG pipeline over a support knowledge base of 100 brands indexed in Moss, returning cited answers with a live confidence score and retrieval-latency tracking.
- Implemented voucher screenshot upload with Tesseract OCR and automatic clarifying questions for incomplete requests.
- Integrated human handoff through the Intercom REST API, with webhook-driven human replies shown live in the chat.
- Containerized the whole product in one Docker container for Render; 64 pytest tests run on every push via GitHub Actions.

Tech Stack: Python, FastAPI, RAG, Moss, Tesseract OCR, Intercom REST API, Docker, Render, pytest

Medical Chatbot using Generative AI
- Built a RAG-based medical chatbot that grounds LLM responses in a retrieved medical knowledge base.
- Designed the retrieval and generation pipeline with LangChain and Pinecone vector search (sentence-transformers embeddings) to supply relevant context per query.
- Developed Flask REST APIs connecting the retrieval layer, Groq-hosted LLM inference, and the chatbot application as a reusable backend service.
- Containerized the application with Docker.

Tech Stack: Python, Flask, LangChain, Pinecone, Groq, sentence-transformers, Docker

Fire & Smoke Detection
- Fine-tuned YOLO11 on a custom-annotated fire and smoke dataset, reaching 92.2% mAP@0.5 (Fire 92.6%, Smoke 91.8%) and an F1-score of 0.89 at a 0.441 confidence threshold.
- Built OpenCV image and video inference with per-frame bounding boxes and confidence scores for continuous monitoring.

Tech Stack: Python, YOLO11, OpenCV

Additional projects on GitHub

- Autonomous Lead Enrichment Agent; Book Recommendation System; Source-Code Analysis tool using GenAI.

WORK EXPERIENCE

ResourcePro, Associate, SDU 06/2025 – 05/2026 | Bengaluru

- Analyzed and validated business data while ensuring high accuracy and quality.
- Followed SOPs and established workflows to process tasks efficiently.
- Investigated data discrepancies and supported issue resolution through detailed analysis.
- Prepared reports and collaborated with stakeholders to support business operations.
- Contributed to process improvements that enhanced operational efficiency and productivity.

AI4SEE Pvt Ltd, Data Analyst Intern 10/2024 – 01/2025 | Bengaluru

- Collected, augmented and cleaned image datasets and annotated them in Roboflow to train YOLOv10 object-detection models.
- Developed a real-time object-detection system integrating YOLOv10 and DETR models, improving detection accuracy with Non-Maximum Suppression (NMS) and Intersection over Union (IoU).
- Contributed to a research paper on garbage classification (biodegradable and non-biodegradable waste) for waste-management automation.
- Worked as the project lead for the research work, taking project-status updates from teammates and reviewing their progress.
- Worked on Graph Neural Networks (GNNs) for traffic prediction, improving real-time traffic-forecasting accuracy.

iNeuron Intelligence Pvt Ltd, Data Scientist Intern 07/2024 – 10/2024 | Bengaluru

- Developed an LLM-powered data analysis system using the Groq API, automating insight generation.
- Implemented real-time processing and predictive modeling, adapting to changing data patterns.
- Designed scalable, secure data pipelines supporting business intelligence applications.
- Explored and compared the performance of various LLMs for the data analysis tasks.

EDUCATION

Harsha Institute of Management Studies, Bangalore University, MCA 2022 – 2024 | Bengaluru
First Class Exemplary, CGPA 8.22

Sahyadri Science College, Kuvempu University, BCA 2019 – 2022 | Shivamogga
First Class, 74.5%

CERTIFICATIONS

1. Fundamentals of Generative AI - Microsoft
2. Artificial Intelligence - IBM
3. Generative AI Mastery & Full Stack Data Science - Euron
4. Soft Skills & Personality Development - CRACK-ED